

Review

Using the gradient method to determine soil gas flux: A review



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ABSTRACT

Gas exchange between soil and atmosphere represents a major component of global greenhouse gas fluxes. Chamber methods and micro-meteorological methods are well-established techniques to measure gas fluxes. The gradient method is not as widely used, but it has gained increased attention during the last decade. In this review we provide an overview of the gradient method, from the concept over different aspects of the application to the limitations and challenges of the method.

Assuming gas diffusion as the dominant transport mechanism, gas flux in porous media such as soil or snow can be calculated based on the profiles of gas concentrations and soil gas diffusivity. A variety of systems has been used to determine the vertical gas profile depending on the objective of the respective study. The estimation of soil gas diffusivity is a major source of uncertainty. Soil gas diffusivity can be derived using diffusivity models, laboratory measurements or *in situ* approaches, e.g. the Radon method. Choosing a diffusivity model has to be considered carefully, since flux estimates are directly affected. Different approaches to calculate the gas flux have been introduced, from direct simple calculations to analytical and numerical solutions. Flux estimation is highly sensitive to the calculation procedure. It is important therefore to consider the implicit assumptions of each calculation approach.

Several studies compared flux estimates measured by the gradient method and other methods. Good agreement was found in studies of CO₂ production and methane consumption in soils, particularly in studies using near-continuous measurements of CO₂. The relation was not as strong for fluxes of CH₄ and N₂O. Deviations were attributed to the possible coexistence of production and consumption of methane and N₂O in the top soil layer.

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1. Introduction

Interest in soil processes in recent decades has led to numerous and diverse studies focused on the measurement of soil–atmosphere gas exchange. Greenhouse gases like CO₂, CH₄, and N₂O in particular have received much attention because of the prominent role in the discussion about global climate change (Forster et al., 2007). The most important field methods used in this context are the micro-meteorological eddy covariance method (Baldocchi et al., 1988; Massman and Lee, 2002), chamber methods (Davidson et al., 2002; Lundegardh, 1927), and the gradient method (De Jong and Schappert, 1972; Schack-Kirchner et al., 2011). Each of these approaches differs greatly in scale of observation, assumptions, limitations and applicability.

The eddy covariance method estimates turbulent gas transport, which dominates gas transport in the atmosphere above the soil most of the time. Yet, it cannot be used when turbulence is not sufficient, i.e. in calm situations, which can occur quite often e.g. during nights or below dense plant canopies. Eddy covariance measurements estimate ecosystem net flux, meaning they measure not only soil–atmosphere exchange but also plant-derived fluxes such as plant photosynthesis during the day and plant respiration during the night. The footprint area of the measurement depends on the wind situation and can cover up to hundreds of m². This allows good overall flux estimation at a site, yet it hampers a detailed investigation of the spatial variability of soil processes. Investigating this variability is essential when we want to understand the underlying soil processes.

Open-bottom chamber methods are widely used to quantify soil gas fluxes. There are different types of chambers (Janssens et al., 2000; Pumpanen et al., 2004), e.g. steady-state chambers (SSC) and non-steady-state chambers (NSSC). Yet, all chamber methods alter the gas environment – the soil–atmosphere gradient and, thus, the pre-deployment soil–atmosphere flux – that they intend to measure (Hutchinson et al., 2000). A few studies modeled gas transport in soil-chamber systems to investigate the effect of chamber deployment and test different flux calculation approaches (Hutchinson et al., 2000; Livingston et al., 2006; Sahoo and Mayya, 2010). Which calculation approach to follow to retrieve the pre-deployment flux, however, is still controversially discussed (Kutzbach et al., 2007; Jassal et al., 2012; Sahoo and Mayya, 2010). Chamber methodology can differ in many more characteristics (Rochette and Eriksen-Hamel, 2008; Pihlatie et al., 2013). The insertion of collars into the soil can also significantly affect the soil respiration (Heinemeyer et al., 2011). Additionally, chambers can affect the micro-climate, which can, in turn, alter the production or consumption of gas. For automated measurements, the exclusion or modification of rainfall and litter fall can be significant.

The gradient method (GM) is based on the measurement of transport-driving gas concentrations gradients and transport properties. Based on the assumption that molecular diffusion dominates gas transport in soils, gas fluxes can be calculated from the gradient of soil gas concentration and the effective gas diffusivity of the porous medium. While eddy covariance and chamber methods only measure soil–atmosphere gas exchange, the GM can provide additional information about the depth profile of net gas

production or consumption in the soil. Further, problems associated with the use of chambers, like disturbance of the concentration gradient between soil and atmosphere, changes in the chamber environment and micro-climate, can be reduced or avoided.

De Jong and Schappert (1972) were the first to describe the application of the GM in detail, although other researchers used the concept independently (e.g. Albrecht et al., 1970, and Burford and Stefanson, 1973, for N₂O; Clements and Wilkening, 1974, for ²²²Rn). In some studies, chamber measurements were combined with soil gas measurements to partition vertically the soil gas flux using the GM (Davidson and Trumbore, 1995; Sotta et al., 2007). Comparisons of soil–atmosphere fluxes derived using the GM and chamber methods generally showed good agreement, depending on the gas species and other factors. Non-destructive, vertical partitioning of soil gas fluxes is a unique feature of the GM. Thus, comparisons of methods focus on soil–atmosphere flux estimates.

In the past decade, new sensors and measurement devices have been developed. This development enabled and stimulated the use of the GM for continuous soil gas flux studies (Flechard et al., 2007; Hirano et al., 2003; Tang et al., 2003; Vargas et al., 2010). For instance, Tang et al. (2005b) combined continuous soil CO₂ efflux estimates derived using the GM to cover temporal variability of the soil gas flux and periodical chamber measurements to cover the spatial variability. These studies demonstrated the GM's potential as a suitable tool for both short and long term studies. As a consequence, the method has gained increased attention and has been applied in several studies of soil gas exchange.

Many researchers used the GM to study soil respiration (e.g. Davidson et al., 2006; De Jong and Schappert, 1972; Schack-Kirchner et al., 2011). Soil respiration includes autotrophic and heterotrophic respiration which differs in their isotopic CO₂ signatures (Trumbore, 2006). The isotopic CO₂ composition of soil air can help to elucidate the carbon allocation in the soil (Brüggemann et al., 2011; Hanson et al., 2000). Hence, the isotopic analysis of CO₂ has been included in several studies using the GM (¹⁴CO₂: Gaudinski et al., 2000; Hirsch et al., 2002; Trumbore et al., 2006; Winston et al., 1997; ¹³CO₂: Amundson and Davidson, 1990; Amundson et al., 1998; Bowling and Massman, 2011; Goffin et al., in press; Kayler et al., 2008).

In ecological studies, the GM has also been used to investigate soil gas fluxes of nitric oxide, (Gut et al., 2002), N₂O (e.g. Burford and Stefanson, 1973; Massman et al., 1997; Pihlatie et al., 2007) and CH₄ (e.g. Dörr et al., 1993; Dunfield et al., 1995; Striegl et al., 1992; Whalen et al., 1992; Wolf et al., 2011). Due to similarities with soil, the method has also been applied to measure gas fluxes in snow (Brooks et al., 1997; Hubbard et al., 2005; Mast et al., 1998; Monson et al., 2006; Schindlbacher et al., 2007; Sommerfeld et al., 1993; Zimov et al., 1996).

The GM has been used in technical applications as well, e.g. to survey methane fluxes in landfills (Bogner et al., 1995, 1997), to investigate gas transport of CO₂ in the vadose zone down to 36 m depth (Wood et al., 1993; Wood and Petraitis, 1984), and to model oxygen transport in sulfur blocks and sulfidic waste (Birkham et al., 2010; Elberling, 2005). It was used to monitor the biodegradation of hydrocarbon in the soil (Lahvis and Baehr, 1996), volatile contaminants (Rivett, 1995; Rivett et al., 2011; Werner et al., 2005), and other volatile organic and inorganic gases (Helmig et al., 2009).

The objective of this paper is to give an overview and evaluation of the different GM approaches and its application with focus on ecological studies in soil and practical issues. In this paper we will give (1) a short theoretical introduction of the GM and its basic assumptions, followed by a review of (2) various GM systems, (3) the determination of soil-gas diffusivity, (4) different calculation schemes, and (5) studies comparing the GM and other methods of measuring soil-atmosphere exchange. We also identify (6) limitations and challenges of this approach and offer (7) recommendations for possible future users.

2. The gradient method

Molecular diffusion is the dominating gas transport process in the soil. Formally, this means that the gas transport in porous media such as snow or soil can be approximated by Fick's first law in one dimension,

$$F(z) = -D_s \frac{dC}{dz} \quad (1)$$

where F is the gas flux ($\mu\text{mol m}^{-2} \text{s}^{-1}$), D_s the effective gas diffusion coefficient of the gas species in the soil or snow ($\text{m}^2 \text{s}^{-1}$), C the trace gas concentration ($\mu\text{mol m}^{-3}$) and z (m) the vertical position. Use of Eq. (1) implies the following preconditions:

- (1) **Fick's law:** Strictly speaking Fick's law is not applicable in multi-component mixtures like soil air (Jaynes and Rogowski, 1983). Instead Maxwell Stephan equations or the Dusty Gas Model should be used. However, the error introduced by Fick's law is minimal when the diffusing gas is at trace gas levels, when the diffusivity of the medium is not too low, and when diffusion fluxes are equimolar (Cannavo et al., 2006; Freijer and Leffelaar, 1996).
- (2) **Gas phase transport:** For most gases, Diffusivity in the gas phase is 10^4 – 10^5 times greater than that in the aqueous phase, thereby diffusion in the soil solution is usually neglected. Chemistry of the soil solution, however, can be important. For instance, the diffusive flux of CO_2 in the aqueous phase is affected by the dissolution in water that, in turn, depends on the pH (Schlotter and Schack-Kirchner, 2013). Jassal et al. (2004) concluded that the contribution of the aqueous phase flux is negligible when the ratio of air-filled porosity and total pore-space is greater than 0.12.
- (3) **Diffusive transport:** It is assumed that gas transport in the soil and across the soil-atmosphere boundary is entirely driven by diffusive transport. i.e. mass flow of air due to pressure gradients or other transport processes like ebullition or plant mediated gas transport is assumed to be negligible.
- (4) **One-dimensional gas transport:** Diffusion proceeds from regions of higher concentration to regions of lower concentration. Assuming horizontal homogeneity in the soil including the diffusivity and the production or consumption of gas allows simplifying a 3D description of gas transport to a 1D description.

If one or more of preconditions (1)–(4) fail, increasing uncertainty has to be considered. For instance: at high soil water contents, diffusion in the aqueous phase and even ebullition can become relevant and respectively the dominating gas transport process, resulting in an increasing uncertainty in the flux estimates. In a few studies other diffusion models such as the Dusty Gas Model (Cannavo et al., 2006) or additional transport mechanisms such as advection and dispersion were included to model gas transport more realistically (Birkham et al., 2010; Elberling et al., 1998; Massman, 2006; Massman et al., 1997; Poulsen and Moldrup, 2006). The focus of this paper, however, is on the simple gradient method based only on Fick's law.

Given the preconditions (1–4) can be assumed, the net production or consumption profile of a gas can be derived by considering the equation for conservation of matter.

$$\frac{\partial C_{\text{total}}}{\partial t} = -\frac{\partial F}{\partial z} + \alpha \quad (2)$$

where C_{total} corresponds to the amount of gas per soil volume, t is time, and α the production or consumption rate of the gas. C_{total} includes the amount of gas in both the water-filled and air-filled pores. Depending on the gas and the chemistry of the soil solution, the amount of gas stored in the aqueous phase can be neglected or not (e.g. CO_2 in a carbonated soil solution). In most studies diffusion is assumed to be at quasi steady-state, so that the storage change can be neglected ($\partial C_{\text{total}}/\partial t = 0$). Combining Eq. (1) and (2) yields

$$\alpha = \frac{d}{dz} \left(-D_s \frac{dC}{dz} \right) = -\frac{dD_s}{dz} \frac{dC}{dz} - D_s \frac{d^2 C}{dz^2} \quad (3)$$

As is evident, therefore, α depends not only on the curvature of the concentration profile but also on the gradient of the diffusivity profile (Eq. (3)). Fig. 1 depicts the theoretical concept of the GM. The vertical flux profile (Fig. 1, right) can be easily derived, if the profile of soil gas diffusivity (Fig. 1, left) and the concentration profile of the respective gas species (Fig. 1, middle) is known.

3. Soil gas sampling systems

Information about the soil gas concentrations and soil gas diffusivity is required to apply the GM. Researchers developed several approaches to measure gas concentrations in soil and snow, depending on the respective research question. The different systems can be classified into systems for active gas sampling and systems for passive gas sampling.

3.1. Systems for active gas sampling

Systems that draw air actively from the air-filled pore-space are categorized as active sampling systems (Fig. 2 left). The tubes through which the soil air is drawn can be installed directly from the soil surface (e.g. Albrecht et al., 1970; Sanderman and Amundson, 2010; Whalen et al., 1992), buried at a certain depth (Cerling et al., 1991; Cook et al., 1998) or installed into the wall of a soil pit (e.g. Bekele et al., 2007; Davidson et al., 2006; Davidson and Trumbore, 1995; Fierer et al., 2005; Gaudinski et al., 2000). Most pits are refilled after installation and the natural soil layer structure are restored to minimize disturbance of the soil gas environment (Fierer et al., 2005; Gaudinski et al., 2000; Fig. 2e). In other studies the installation pit are not refilled (Davidson et al., 2004; Davidson and Trumbore, 1995; Schwendenmann et al., 2003; Sotta et al., 2007; Terhune and Harden, 1991, Fig. 2a) and horizontal concentration gradients perpendicular to the pit walls have to be considered. Davidson and Trumbore (1995) addressed this issue as “pit wall effect” and used long tubes to access soil compartments that presumably were only little affected. Yet, other effects like enhanced oxygen supply of deeper soil compartments adjacent to the pit and the stimulation of root growth etc. might be relevant.

Sampling tubes often have a perforated lower end and are made of metal (Dörr et al., 1993; Dunfield et al., 1995; Johnson et al., 1994; Yavitt et al., 1995) or plastic (Billings et al., 1998; Castelle and Galloway, 1990; Fernandez and Kosian, 1987, Fig. 2c). Use of multiple sampling tubes in one shaft allows collecting samples from different depths at a single location while minimizing disturbance (Burton and Beauchamp, 1994; Risk, 2002a; van Bochove et al., 1996, Fig. 2d). While most of the systems are installed permanently for repeated measurements, mobile sampling devices can also be used to assess spatial variability. Richter (1972) used a robust steel probe that was inserted successively deeper into the

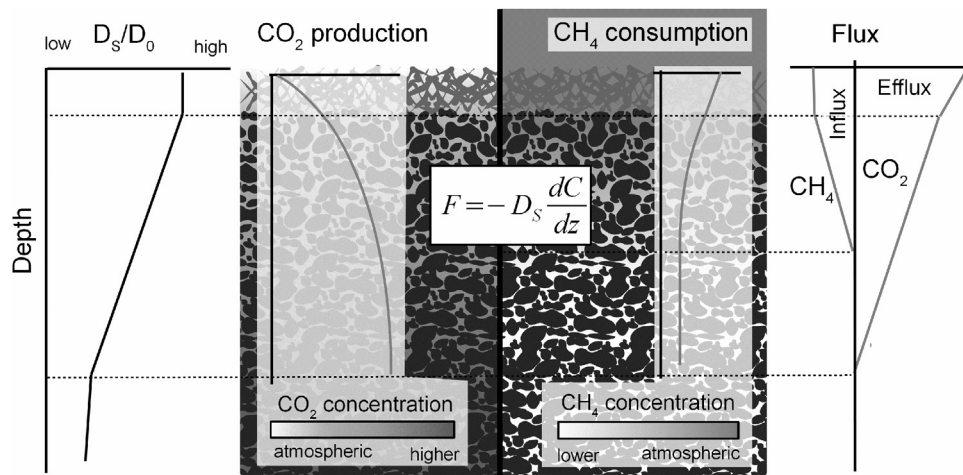


Fig. 1. CO₂ production and CH₄ consumption in a soil profile, where application of the gradient method is ideal. Soil gas diffusivity (left), concentrations of CO₂ (middle-left) and CH₄ (middle-right) are horizontally homogeneous, and fluxes of CO₂ and CH₄ (right) can be derived.

soil to sample air at different depths, moving from location to location. Schindlbacher et al. (2007) and Sullivan (2010) used a similar approach in snow. In all cases, the sampling depth has to be known, which is usually measured as depth below the soil–atmosphere boundary. In snow, however, the changing thickness of the snow cover has to be considered, e.g. by installing gas sampling towers with ports at different heights before the onset of snowfall (Bowling et al., 2009; Filippa et al., 2009; Kim et al., 2007; Mast et al., 1998; Sommerfeld et al., 1996; Takagi et al., 2005, Fig. 2b).

A significant vulnerability of active gas sampling is that it alters the concentration gradients of the gas environment in the soil or snow pores. The affected volume depends on the air-filled porosity, the size of the required gas sample and the change in pressure used to extract the sample. The required sample size depends on the type of gas analyzer, and can vary from a few ml (e.g. Davidson et al., 2004) to liters (Musselman et al., 2005; Seok et al., 2009). As a consequence the depth from which the sample is drawn may be uncertain. Further, measurements cannot be made continuously, and a minimum distance between sampling depth and locations is required. In most cases, active sampling is used in studies with a low temporal resolution, e.g. with the focus on seasonal effects.

3.2. Systems for passive gas sampling

In contrast to active gas sampling systems, soil trace gas concentrations in passive systems are measured using a sampling tube

or sensor that is allowed to equilibrate with the soil air *in situ* (Fig. 2 middle, right). For example, Taylor and Abrahams (1953) connected a vial to a sampling tube inserted at depth over an extended period to allow the gas in the vial to equilibrate with that in the soil pores. The sample vial was then removed and analyzed. Similarly, Butterbach-Bahl and Papen (2002) used soil gas probes protected by a Teflon membrane. The air was then actively sampled from the equilibrated probes. Using a passive sampling system, it is possible to limit disturbances to the natural soil gas environment, ideally reduced to the initial installation of the system. This also allows for fine vertical resolution sampling. Schack-Kirchner et al. (1993) used a similar approach to investigate the concentrations gradient in the litter layer (Fig. 2f). Kammann et al. (2001) used closed silicone tubing inserted at depth to sample soil gas in wet soils (Fig. 2g). Water could not penetrate through the tube wall, but gases diffused rather quickly through the silicone. When it was assumed that gas concentrations in the silicone were in equilibrium with that in the soil, the air from the tubing was sampled using a syringe. The described passive sampling systems require manual sampling and the sampling interval is usually on the order of days to weeks to allow for equilibrium.

Passive gas sampling can also be combined with automated gas analysis to obtain (nearly) continuous measurements. Gut et al. (1998) used gas permeable, hydrophobic polypropylene tubes to sample gas at different soil depths, but other materials can be used as well (DeSutter et al., 2006). Gut et al. (1999) employed a

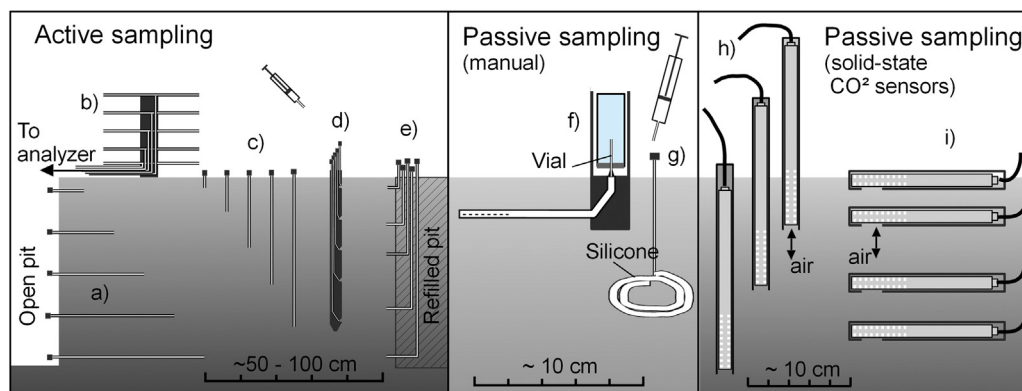


Fig. 2. Soil gas sampling systems. Left: Active sampling systems. (a) Sampling tubes in an open pit, (b) snowpack sampling tower, (c) gas wells, (d) multiple sampling tube, (e) sampling tubes in a refilled pit. Middle: Passive sampling (manual). (f) Diffusion-equilibrium sampler, (g) “closed silicone tube” sampler. Right: Passive sampling using solid-state CO₂ sensors. (h) Vertical installation, (i) horizontal installation.

closed-loop sampling system within which the air was circulated from the sampling tube through the gas analyzer and back into the tube. The sampling tubes in this system stretched a distance of >1 m in the soil, therefore averaging concentrations over a variety of soil micro-environments. The sampling tubes can easily be installed in soils used for agriculture by burying them at the defined depth (DeSutter et al., 2008; Flechard et al., 2007; Gut et al., 1998). Installation in an undisturbed natural soil, however, requires a more careful procedure (Parent et al., 2013; Wolf et al., 2011). Albanito et al. (2009) used sampling chambers instead of tubes in a closed loop sampling approach. The gas permeable section of the chamber had a height of 5 cm however, making it difficult to relate the concentration measurement to a precise depth.

The development of small solid-state CO₂ sensors has greatly facilitated CO₂ concentration measurements in the soil over the last decade (Vargas et al., 2010). The sensors can be either installed into the wall of pits that are refilled (Daly et al., 2009; Hirano, 2005; Jassal et al., 2005; Maier et al., 2010; Pumpanen et al., 2008; Turcu et al., 2005, Fig. 2i) or installed vertically into inert sampling tubes from the soil surface (Chen et al., 2005; Sullivan et al., 2010; Tang et al., 2003; Tang et al., 2005a,b; Vargas and Allen, 2008a,b,c; Young et al., 2009, Fig. 2h). Hydrophobic gas-permeable caps around the sensors allow the soil air to equilibrate with the air inside the measurement chamber of the sensors. This development has made near-continuous soil CO₂ measurements possible, thus offering the opportunity for studies on very short time-scales (Baldocchi et al., 2006; Hasselquist et al., 2010; Tang et al., 2005a; Vargas et al., 2011). Continuous operation of the sensors, however, can heat up the soil, meaning intermittent operation of the sensors is recommended (Hirano et al., 2003; Jassal et al., 2005).

An important issue for all systems is uncertainty in the sampling position. Errors in the measurement depth directly affect the flux calculation. To assess the concentration gradient at the surface, measurements as close as possible to the surface are needed, e.g. at 2 cm depth (Tang et al., 2003; Vargas and Allen, 2008a). At such shallow depths, errors in the depth estimate as small as 0.5 cm can lead to relatively large errors in the resultant concentration gradient estimates (Hirano et al., 2003). Precise determination of the installation depth, however, can be difficult in practice due to surface roughness. Further, the thickness of the litter layer can vary horizontally and temporally (litter fall, decomposition etc.). Wind can penetrate into the uppermost macro-pores (Novak et al., 2000a, 2000b) so that effective thickness of the litter layer also depends on the wind situation. The determination of the sampling depth is also an issue when long, gas permeable sampling tubes are used because the exact distance to the surface may vary along the length of the tube (DeSutter et al., 2008; Wolf et al., 2011).

If soil gas flux is investigated on shorter time scales, e.g. with the focus on intra-diurnal changes, concentration changes in the atmosphere can be relevant. CO₂ concentrations near the soil surface can be highly variable (Hirsch et al., 2004; Maier et al., 2010). The influence on the flux calculations can range from low in soils with steep concentration gradients to high in soils where gradients are shallow. Thus, the atmospheric concentration near the soil surface should be monitored.

4. Determination of the diffusivity profile

A major source of uncertainty in the use of the gradient method since its introduction is estimation of the effective diffusivity D_S (Davidson and Trumbore, 1995; De Jong and Schappert, 1972; Koehler et al., 2010; Pingintha et al., 2010; Schwendenmann and Veldkamp, 2006). D_S in porous media depends on the intrinsic properties of the diffusing gas (diffusivity in free air D_0) and on the air-filled pore structure (air-filled pore-space, tortuosity), which, in

turn, depends on the soil moisture status. D_S can either be modeled or measured directly *in situ*.

4.1. Transfer models for soil gas diffusivity

The most common approach is to use models that derive the dimensionless relative diffusivity D_S/D_0 from the air-filled pore-space. The air-filled pore-space can be determined as the difference between the total pore-volume and the soil water content. The spatial variability of total pore-volume should be considered. Total pore-volume can be assumed to be constant with time in most cases, while soil water content changes over time. Thus, D_S is most influenced by changes in soil water content during a monitoring campaign. A soil-specific calibration of soil moisture probes is recommended, since their output directly affects the calculation of the diffusivity and the flux.

There are many models of differing complexity that link soil D_S/D_0 and air-filled pore-space (see e.g. Allaire et al., 2008). Some models use only a few simple parameters like air-filled pore-space and/or total pore-space as inputs (Marshall, 1959; Millington and Quirk, 1961; Penman, 1940). Other models require additional information about the pore size distribution and water retention (Millington and Shearer, 1971; Moldrup et al., 2000, 2005).

Nevertheless, these generalized models cannot account for site specific characteristics. This problem can be overcome by measuring D_S of undisturbed soil samples from the respective site (Goffin et al., in press; Jassal et al., 2005; Koide et al., 2010; Maier et al., 2010; Pihlatie et al., 2007; Pumpanen et al., 2008; Wolf et al., 2011). Measurements should cover the range of expected soil moisture and include several soil samples per layer to account for spatial variability. Non-linear functions (e.g. Currie, 1960; Troeh et al., 1982) can be fit to the measured D_S data to obtain a site-specific D_S model.

To apply the GM in snow layers, D_S of snow is required. As with soil, both the snow pack porosity (Alm et al., 1999; Maljanen et al., 2003; Pihlatie et al., 2007; Rochette et al., 2000) and the tortuosity have to be considered to calculate D_S/D_0 (Mast et al., 1998; McDowell et al., 2000; Winston et al., 1995).

Soil gas diffusivity (D_S) can be calculated by multiplying D_S/D_0 by D_0 . Molecular diffusivity of a gas is influenced by air temperature T expressed in K and barometric pressure p , so that these effects should be considered using e.g. Eq. (4):

$$D_0(T, p) = D_0^{\text{ref}} \left(\frac{p_{\text{ref}}}{p} \right) \left(\frac{T}{T_{\text{ref}}} \right)^{\alpha} \quad (4)$$

where D_0^{ref} is the D_0 at reference barometric pressure p_{ref} and reference air temperature T_{ref} , and α is a gas specific exponent between 1.5 and 2 (Massman, 1998).

4.2. Limitations and challenges in the application of diffusivity models

A continuous profile of diffusivity is needed to determine the production profile (Eq. (3)). If soil moisture is measured at discrete locations, it is necessary to interpolate data. Linear interpolation of the D_S values between the measurement locations is the simplest method, but not always realistic. For instance, soil water content can be homogenous within soil layers yet change substantially at the boundary of soil layers with distinct soil water retention. Even homogeneous soils can exhibit a non-trivial, transient soil water profile after rainfalls. The propagation of the wetting front between the measurement depths of the soil moisture can be reproduced by use of a soil water flow model (Kato et al., 2012; Turcu et al., 2005). The vertical distribution of the soil water content has to be considered to calculate a realistic D_S . It is important to note, that the harmonic mean must be used to calculate the resulting D_S of the

layer between the gas concentration measurements (Turcu et al., 2005).

Many soils are covered by a litter layer with a high porosity. The litter layer is subject to fast changes in temperature and moisture, and even the thickness can change with time. To our knowledge only scarce data about the diffusivity of litter exists (Maier et al., 2012; Schack-Kirchner et al., 2011), and common D_s models can hardly be used. Gas transport through a highly porous layer can be significantly affected by wind (Hirsch et al., 2004). Firstly, the near surface concentration may underlie diurnal cycles that are triggered by wind (e.g. Maier et al., 2010). Secondly, movement of air itself can penetrate into the macro-pores and cavities in the litter layer (Novak et al., 2000b, 2000a) and shift the effective atmosphere–soil boundary. Thirdly, turbulence-driven pressure-pumping can enhance soil gas transport above the level of molecular diffusion (Kimball and Lemon, 1971a; Maier et al., 2012; Scotter and Raats, 1968).

The estimation of D_s is particularly challenging in soils with macroscopic features, such as rocks or cracks. Macroscopic cracks in a drying clay soil or rock can substantially increase gas exchange between soil and atmosphere (De Jong and Schappert, 1972; Weisbrod et al., 2009). High rock content might also cause deviations between modeled D_s values and D_s values estimated using an *in situ* method to determine D_s (Davidson et al. (2006)). Even site specific D_s models derived from D_s measurements can hardly take these macroscopic features into account since such measurements are usually conducted on smaller soil samples (e.g. Kusa et al., 2010). An approach to overcome these problems is to measure soil gas diffusivity *in situ*.

4.3. *In situ* measurement of gas diffusivity

Soil gas diffusivity can be measured *in situ* using a tracer gas. The most important *in situ* method is the radon (Rn) method (Davidson and Trumbore, 1995; Dörr et al., 1993; Dörr and Münnich, 1990; Fierer et al., 2005; Koehler et al., 2010; Ota and Yamazawa, 2010; Schwendenmann and Veldkamp, 2006; Sotta et al., 2007; Uchida et al., 1997; Whalen et al., 1992). This method takes advantage of the ubiquitous, naturally occurring, radioactive noble gas ^{222}Rn . Radon is produced by the alpha-decay of radium, a part of some trace minerals in the soil. Assuming diffusion as the transport mechanism and accounting for the radioactive half-life of ^{222}Rn , D_s can be derived (Dörr and Münnich, 1990). The profile of the radon production rate must be measured from samples at different depths too (e.g. Davidson and Trumbore, 1995; Fierer et al., 2005). In general, a good agreement between the Rn derived D_s and a model derived D_s can be found (or at least a D_s model that fits to the Rn derived D_s).

Artificial sources of tracer gases can also be used to measure soil gas diffusivity *in situ*. Werner et al. (2004) classified the different set-ups in a review into flux chamber methods (e.g. Ball et al., 1994; McIntyre and Philip, 1964; Schack-Kirchner et al., 2001), methods using instantaneous point sources (e.g. Hers et al., 2000; Johnson et al., 1998; Lai et al., 1976) and methods using continuous sources (e.g. Kimball and Lemon, 1971a; van Bochove et al., 1998). Risk et al. (2008) modeled diffusivity based on temporary dynamics of CO_2 withdrawal and natural replenishment.

5. Flux calculation approaches

De Jong and Schappert (1972) used the GM to estimate both the soil atmosphere flux and the vertical partitioning of the soil gas flux. They divided the soil into layers limited by the depths of the measurement locations. The flux through each layer was calculated from the D_s and the discrete difference in the gas concentration

across each layer. The flux was assigned to the center of the layers (Fig. 3; F_a, F_b, F_c). The production or consumption was calculated as the change in the flux from the center of one layer to the center of the next layer. This approach has been widely used and sometimes slightly modified, for example by shifting the flux through the top soil layer to the soil surface (Fig. 3 $F_a \rightarrow F_0^1$, Maier et al., 2010; Wolf et al., 2011).

5.1. Estimation of the soil–atmosphere flux

Many studies used the GM to estimate the soil surface flux (F_0) to complement or replace chamber measurements. To estimate precisely the soil–atmosphere flux, theoretically the gas concentration gradient and the effective diffusivity at the soil surface must be known. In practice, the measurement of the gas concentration gradient must be made across a finite soil layer at the soil surface. The production and consumption of gas, and changes in diffusivity can result in a changing gradient across that layer so that the distance between the concentration measurements and the distance to the surface should be as short as possible. Yet, the minimal distance is limited by the sampling system needed to measure soil gas concentrations and the accuracy with which the sampling location can be determined. Depending on the production or consumption of gas in that layer, the layer's thickness and localization, and changes in the concentration gradient and gas flux can or cannot be negligible.

Tang et al. (2003) studied soil respiration in a very dry soil and observed a rather linear increase of CO_2 concentration with depth. Thus, they used linear regression to estimate the gradient at the surface (Fig. 3 F_0^2). Yet, biological activity and production of CO_2 in the top soil can be high, and the concentration gradient can change substantially. Hirano et al. (2003) addressed this problem by assuming a constant CO_2 production profile in the top layer. This approach has also been used by Hirano (2005), Liang et al. (2004) and Liang et al. (2010). They used four concentration measurements (Fig. 3 at depth = 0, z_1, z_2, z_3), and extrapolated the increase in flux between the first two concentration measurements (Fig. 3, F_a) and the last two concentration measurements (Fig. 3, F_c) to the soil surface (Fig. 3, $F_c, F_a \rightarrow F_0^3$) by using:

$$F_0^3 = \frac{z_c F_a - z_a F_c}{z_c - z_a} \quad (5)$$

Tang et al. (2005b) followed a similar approach which was later applied by Baldocchi et al. (2006) and Vargas and Allen (2008a, 2008b, 2008c). To avoid the difficult estimation of D_s in the top few cm of soil and litter, they excluded the concentration measurement at the soil surface. Three gas concentration measurements were used to calculate the flux between the respective measurement locations and then extrapolate to the soil surface (Fig. 3, $F_c, F_b \rightarrow F_0^4$):

$$F_0^4 = \frac{z_c F_b - z_b F_c}{z_c - z_b} \quad (6)$$

Both approaches of Hirano et al. (2003) and Tang et al. (2005b) showed good agreement in their comparisons of soil–atmosphere flux estimates derived using the GM and flux derived using chamber methods (Baldocchi et al., 2006; Liang et al., 2004). However, using different calculation approaches can result in substantially different flux rates (Fig. 3). DeSutter et al. (2008) compared different approaches to chamber derived flux estimates. The power function approach of Gaudinski et al. (2000), though originally not used to estimate the soil–atmosphere flux, overestimated the surface flux by several orders of magnitude. The linear regression approach of Tang et al. (2003), which was originally used in a soil with inactive topsoil, substantially underestimated the measured chamber flux, and the local linear approach of De Jong and Schappert (1972) was close to the chamber derived flux.

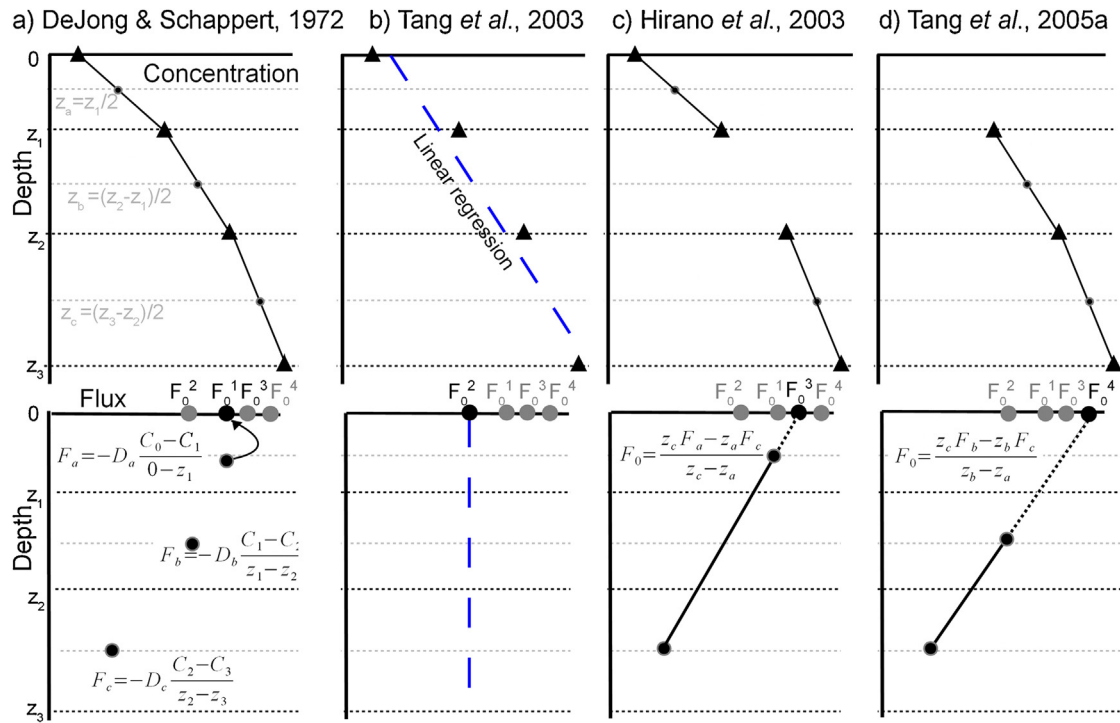


Fig. 3. Different calculation approaches to derive the soil-atmosphere flux (F_0) using the GM. The upper panels depict a soil profile with concentration measurements (triangles). The lower panels depict the flux profile. The gradient (black and gray circle) is assigned to the position between the measurement locations in the upper panel, and is used to calculate the flux at the respective depth (black and gray circle) displayed in the lower panels. F_0 is calculated based on (a) discrete differences^c, F_a is shifted to F_0 , (b) linear regression^f (dashed line), or (c, d) extrapolated fluxes^{g,h} (the dotted lines represent the extrapolation). F_0 calculated using the reference approaches are kept as gray circle in the respective panel.

5.2. Vertical partitioning of soil gas fluxes

The production or consumption of a gas in a soil layer can be calculated as the difference between the flux at the lower boundary of the soil layer and flux at the upper boundary, if the presumption of quasi steady-state diffusion is true (De Jong and Schappert, 1972). To estimate the soil-atmosphere gas flux using the GM, knowledge of the soil gas concentration and D_s near the surface is needed. To circumvent this, some researchers combined the profile measurements of the GM with chamber measurements to derive the soil atmosphere flux (Davidson et al., 2004, 2006; Davidson and Trumbore, 1995; Hashimoto et al., 2007; Koide et al., 2010; Schwendenmann and Veldkamp, 2006; Sotta et al., 2007). The production in the deeper layers was calculated as the difference between the influx and efflux from the respective layer using the GM. The production of the top layer was estimated as the difference between the chamber flux and the influx at a certain depth determined by the GM (Davidson and Trumbore, 1995).

An alternative approach to calculate discrete fluxes between measurement locations is to fit an analytical function to the discrete concentration measurements to yield continuous, differentiable, vertical profiles of soil gas concentrations (Fig. 4). The derived concentration profile and the D_s profile allow the flux profile to be easily derived. Following this empirical approach, different analytical functions were fitted to a virtual profile of discrete concentration measurements (Fig. 4). Gaudinski et al. (2000) and Davidson et al. (2006) used an exponential function (Eq. (7)), where the fit parameter C_∞ represented the concentration in the deep soil, 0.04 was the atmospheric CO_2 concentration and β was a shape factor:

$$C(z) = C_\infty(1 - e^{\beta z}) + 0.04 \quad (7)$$

Jassal et al. (2005), Takle et al. (2004), Daly et al. (2009) and Koide et al. (2010) fitted a quadratic function (Eq. (8)) to their data to derive a continuous CO_2 profile, where a , b and c are fit parameters:

$$C(z) = az^2 + bz + c \quad (8)$$

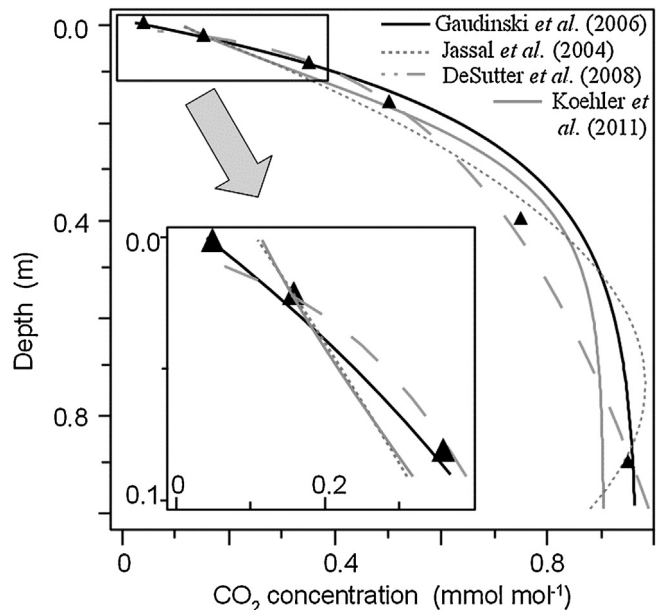


Fig. 4. Fitting of different analytical functions to discrete (simulated) concentration measurements (triangles) to estimate the continuous concentration depth profile. Note that the best-fit function for the whole profile does not necessarily yield the best representation of the gradient next to the soil surface (enclosure).

DeSutter et al. (2008) tested a power function (Eq. (9)) to fit experimental soil CO₂ data, where d , e and f are fit parameters:

$$C(z) = d + ez^f \quad (9)$$

The concentration profile can exhibit an inflection point that cannot be represented in exponential or quadratic functions (Albertsen, 1979). Koehler et al. (2010) used the asymmetric sigmoidal Gompertz function (Eq. (10), Richards, 1959) to include inflection points, where g , h and i are fit parameters:

$$C(z) = ge^{he^{iz}} \quad (10)$$

Applying one of the functions above (Eq. (7)–(10)) fits or shifts the concentration profile toward the assumed curvature of the applied fitting function. The best-fit function for the whole profile does not necessarily yield the best fit of the gradient at the soil–atmosphere boundary (Fig. 4). Fitting the exponential function (Gaudinski et al., 2000) or the power function (DeSutter et al., 2008) to soil CO₂ data can yield an unrealistically high efflux at the surface due to the increasing steepness of the function toward the soil surface (Schack-Kirchner et al., 2011; DeSutter et al., 2008). Numerical modeling of soil gas transport shows that gas concentration gradients do not necessarily increase as they approach the soil surface, since the simultaneous increase in D_s compensates this development (Schack-Kirchner et al., 2011; Jassal and Black, 2006). Using the Gompertz function (Eq. (9)) for this purpose, on the other hand, can underestimate a flux when compared to chamber derived estimates (Koehler et al., 2010). Which function yields the best fit (e.g. measured as R^2) depends on the measured concentrations and the number and location of those measurements (Fig. 4). If the shape of the concentration profile changes, e.g. due to changes in diffusivity after rain (see Fig. 9 in Yoshikawa and Hasegawa, 2000), another function might fit better. Hence the choice is somewhat arbitrary, yet strongly affects the outcome of the modeling.

Exclusively fitting empirically analytical functions to measured soil gas concentration profiles is straightforward, yet it lacks physical background. The profile of soil gas concentrations depends not only on the profile of gas production or consumption, but also on the D_s profile. Hence, information about D_s must be included to derive the gas production or consumption profile. Working with virtual data, Drewitt et al. (2005) assumed different production profiles of soil CO₂ and a constant D_s throughout the soil. Based on this, they developed analytical solutions that could be fitted to measured data and numerical solutions that could be inversely fitted. Following a similar approach, Takahashi et al. (2004) included an exponential function to describe the D_s profile. Even though this allows for a better representation of the D_s profile, such an exponential decrease in D_s with depth is not true in some cases, e.g. in profiles with a hardpan (Risk, 2002b), at an ice layer within a snowpack (Schindlbacher et al., 2007) or after rainfall (Turcu et al., 2005).

Novak (2007) tested different inversion techniques to determine the flux profile from concentration data in the soil, and found the inversion errors acceptably low for all techniques. The function used to describe the production profile (e.g. step function vs. exponential function) revealed to be a crucial factor. The distribution of root density and organic carbon content in a soil profile might be related to biological activity and CO₂ production. However, other factors like soil temperature, soil moisture and nutrient availability can change the expected activity. Hence, choosing a function to describe the production profile is a crucial and difficult question.

As stated above the function used to describe the production profile revealed to be a crucial factor (Novak, 2007). This question can be circumvented following an approach where the soil is divided into different independent layers, for which the CO₂ production and D_s is assumed to be homogeneous (Van Bavel, 1951; Schack-Kirchner et al., 2011). Based on these assumptions,

Schack-Kirchner et al. (2011) used quadratic splines to fit the function to the concentration measurements, with the constraint that flux continuity must be adhered to at the spline nodes. Using synthetic data sets, they compared different approaches to calculate the soil–atmosphere flux and the vertical partitioning. The exponential function approach (Gaudinski et al., 2000) yielded far too high soil efflux estimates due to the increasing steepness of the function toward the soil surface; vertical partitioning additionally yielded large deviations. The approach of De Jong and Schappert (1972), and the approach of Schack-Kirchner et al. (2011) yielded realistic results compared to the synthetic input data in most cases.

5.3. Soil gas transport modeling

Another way to gain information about soil processes is to combine measured soil gas concentrations, mechanistic models of soil gas transport and numerical modeling. Numerical modeling of soil gas transport can be a solution if the analytical description of the gas transport process fails or would be too extensive. For instance, transient situations often require numerical modeling of soil gas transport, e.g. after intense rainfall (Yoshikawa and Hasegawa, 2000). Numerical modeling does not require parameterized profiles of gas concentrations or D_s . Further, two (and three) dimensional gas transport can be modeled without restrictions regarding the geometry (Striegl and Ishii, 1989; Weltecke et al., 2012). Hence, modeling of soil gas transport can be an important tool for scenario studies (Drewitt et al., 2005; Jassal and Black, 2006) and optimization of sampling designs (Ohlsson, 2010; Weltecke et al., 2012). If mechanistic models are parameterized independently, measurements (e.g. gas concentrations) can be used to test the model assumptions. This allows, in a further step, for inverse modeling of soil gas transport so that flux estimates can be indirectly derived from gas concentration profiles, producing similar results to the GM.

Mechanistic models differ in complexity, starting with production and transport of CO₂ (Fang and Moncrieff, 1999). Most models additionally include transport of soil water and heat (Šimůnek and Suarez, 1993; Jassal et al., 2004), and more processes like solute transport, multiphase CO₂ transport and plant growth (Suarez and Šimůnek, 1997). The crucial point of mechanistic models is reliable parameterization, which can be challenging (Suarez and Šimůnek, 1993). Hence, a model should be as specific as necessary and as simple as possible. To avoid extensive parameterization, Pumpanen et al. (2003) developed a “simple easily parameterized dynamic model” describing the production and diffusion of CO₂ in the soil, and the response to soil moisture and soil temperature. The model was used for the field studies of Pumpanen et al. (2003), Pihlatie et al. (2007) and Pumpanen et al. (2008). Similarly, Jassal et al. (2004) were able to simulate soil CO₂ profiles and observed CO₂ efflux.

It is important to note the difference between the actual GM and soil gas transport modeling. The GM directly uses profile measurements of gas concentrations to derive soil gas fluxes (e.g. to derive, eventually, a temperature-driven soil respiration model). In contrast, for mechanistic soil gas modeling, the model parameter (e.g. for a temperature-driven soil respiration model) is set, and the simulated gas concentration profile is compared to the observed one. If all relevant processes are represented in the mechanistic model, theoretically the parameter can be optimized so that the observed data (e.g. gas concentration profile) can be approximated by the simulated data. By using optimization algorithms, this can be done automatically (inverse modeling). It is thus possible to inversely fit a model with measured concentration data and D_s so that, similar to the GM, flux data can be derived (indirectly). Programs that allow for inverse modeling are e.g. the Finite Element Model (FEM) programs HYDRUS (Šimůnek et al., 2005) and

COMSOL Multiphysics (COMSOL Inc., Burlington, Massachusetts, US). To date, inverse modeling approaches have been used for soil gas modeling e.g. by Novak (2007), Ohlsson (2010) and Weltecke et al. (2012).

6. Comparison of chamber and gradient method measurements

Several studies compare soil gas flux estimates derived using the GM with those derived using micro-meteorological or chamber methods (Table 1). Confounding the interpretation of these comparisons, however, are the different aspects of the studies such as the sampling locations, the D_s model used, the gas species, the approaches taken to estimate the surface flux for the GM and especially the specific chamber methods.

Soil gas fluxes can exhibit large spatial and temporal variability. The GM measurements and the reference measurements are usually not conducted at the same location simultaneously, nor do they measure the same footprint area. While the mean of several chambers is used as a reference in many cases, the GM is often applied only at a single location. Hence, the observed deviation might be due to spatial variability, not differences in methodology (Tang et al., 2005b; Uchida et al., 1997). Further, even flux estimates derived using different chamber systems may differ substantially (Janssens et al., 2000; Pumpanen et al., 2004).

Although it is difficult to draw a general conclusion from the studies listed in Table 1, most authors find a good agreement between the GM and the reference methods. The agreement seems to be better for CO_2 fluxes than for N_2O or CH_4 fluxes, yet there are only few studies on N_2O and CH_4 . Unfortunately, the studies express the agreement between the methods in terms of different parameters.

Soil gas diffusivity is a crucial parameter for the flux calculation using the GM. The D_s model chosen can largely affect the GM soil–atmosphere flux estimation, yielding higher or lower flux rates than chamber measurements (Pinging et al., 2010; Sullivan et al., 2010; Tang et al., 2003; Wolf et al., 2011). It is important to note that there is no ‘universal best D_s model’ but only the model that best represents the respective soil. Generally, a good agreement between the methods was found in studies where soil-specific D_s models derived from measured D_s data were used (De Jong et al., 1979; Jassal et al., 2005; Kusa et al., 2008; Liang et al., 2004, 2010; Wolf et al., 2011). The weak agreement of the methods observed by Kusa et al. (2008) (Table 1) is difficult to interpret. The non-steady-state chamber method used was based on only two gas samples per flux measurement, which resulted in an undefined uncertainty in the flux estimates. A good agreement was also found in studies using the Rn-calibrated method where soil gas diffusivity was determined *in situ* (Ota and Yamazawa, 2010; Uchida et al., 1997).

Most studies comparing these methods are about CO_2 fluxes and use non-steady-state chambers. Even though a few researchers found the GM to overestimate (Pumpanen et al., 2008; Risk, 2002b; Suwa et al., 2004; Wolf et al., 2011) or underestimate the CO_2 flux measured by non-steady-state chambers (Kusa et al., 2008, Liang et al., 2004; De Jong et al., 1979), most studies observed a satisfactory agreement between the methods (see Table 1). Good agreement was found especially in studies using small solid state CO_2 sensors that were permanently installed in the soil (see Table 1; e.g. Jassal et al., 2005; Tang et al., 2005b; Vargas and Allen, 2008a). These probes have facilitated the investigation of diurnal variations of soil CO_2 concentration. It seems that even if GM derived efflux estimates agree closely to chamber measurements during stable weather conditions, they may deviate during and after rain for a short time (Jassal et al., 2005; Riveros-Iregui et al., 2008; Table 1).

CO_2 soil flux estimates derived using steady-state chambers seem to be lower than those derived using the GM. Yet, this observation is based on only two studies (Liang et al., 2004, 2010). Sullivan et al. (2010) found slightly higher soil CO_2 efflux estimates derived using the GM than estimates of the nighttime ecosystem respiration derived using the eddy covariance method. Theoretically, the flux rate of nighttime ecosystem respiration should be higher than that of soil respiration.

Comparing N_2O and CO_2 fluxes at the same site, Pihlatie et al. (2007) and Wolf et al. (2011) found that GM derived estimates for N_2O fluxes were poor despite good estimates for CO_2 . Similarly, Arah et al. (1991), Maljanen et al. (2003) and Kusa et al. (2008) observed that the chamber-measurements of N_2O fluxes could not be approximated by the GM. soil–atmosphere N_2O flux exhibits higher spatio-temporal variability on a small scale than the flux of CO_2 , challenging assumptions made in the application of the GM and chamber methods. Vertically within the soil profile, production of CO_2 is more or less evenly distributed; production of N_2O , however, often occurs in (anaerobic) micro-sites. If such micro-sites occur in the topsoil above the uppermost gas sampling location, uncertainty in GM derived flux estimates of the soil–atmosphere N_2O flux increases because the surface gradient will be poorly estimated. Thus, soil–atmosphere N_2O fluxes derived using the GM should be carefully considered.

CH_4 is only produced in soils under strictly anaerobic conditions (Peters and Conrad, 1995; Le Mer and Roger, 2001), representing non-equimolar gas turnover. If soils are near water saturation, plant-mediated transport and ebullition can become important. These transport processes are not represented using the GM. Production and oxidation of CH_4 can occur simultaneously in different soil compartments. For example, more than 90% of CH_4 produced in the anaerobic zones of a paddy soil can be re-oxidized in the well-aerated aerobic zones (Le Mer and Roger, 2001), where diffusion is the dominant gas transport process. Dunfield et al. (1995) observed a good agreement between chamber measurements of CH_4 oxidation and GM derived estimates. They found the investigated soil “an ideal, unstructured, vertically homogeneous system in which to study gas diffusion” (Dunfield et al., 1995). Hendriks et al. (2010) investigated CH_4 fluxes at different sites near water saturation, and observed that the GM and chamber method estimates agreed only at sites with low water levels, where diffusion as main gas transport mechanism seemed reasonable. Similarly, Billings et al. (2000) found a close agreement between GM and chamber measurements only at one location, and attributed the deviation at other locations to varying effects of soil moisture on both methanotrophy and methanogenesis. Also in other studies, only a rough relation between chamber measurements and the GM (Whalen et al., 1992; Wolf et al., 2011; Yavitt et al., 1995) has been observed. It seems that the GM yields reliable estimates of CH_4 consumption rates in aerated upland soils. If, however, CH_4 is produced – particularly in the topsoil – results can deviate.

Working with snow-pack, Winston et al. (1995) and Mast et al. (1998) observed large differences in the CO_2 and CH_4 flux estimated using the GM and chamber methods. This was attributed to the presence of melt channels and to lateral diffusion below the chambers, which resulted in unreliable estimates of both the chamber method and the GM. For the growing season, Myklebust et al. (2008) found a good agreement between GM estimates and non-steady-state chamber measurements. Yet, the flux estimates seemed to be uncorrelated when the field was covered with straw and snow (Table 1). Working with snow-pack, Schindlbacher et al. (2007), Mariko et al. (1994) and Kim et al. (2007) found the GM to yield higher CO_2 efflux rates than the chamber method, whereas McDowell et al. (2000) found no difference and Mariko et al. (2000) observed higher fluxes from the chambers. Alm et al. (1999) found a good agreement between the average flux rates of CH_4 and CO_2

Table 1

Accuracy of the GM flux estimates in comparison to reference measurements. Reference measurements were made using alkali absorption chambers (AAC), steady-state chambers (SSC), non-steady-state chambers (NSSC), and the micro-meteorological eddy-covariance (EC) method. Agreement between the methods is given as slope of the regression flux rate GM vs. flux rate reference (a), coefficient of determination (R^2) and relative accuracy (r.a.).

Gas	Site	Gradient method ^a			Reference system (relevant available details)	Relative accuracy	Reference
		Set-up	D_5 profile ^b	Calculation			
CO ₂	Oak-grass savanna (dry)	IRGA installed in soil	M-Q1961 Ma1959	(a) (a)	NSSC, bi-weekly	$a = 0.91$, $R^2 = 0.84$ $a = 1.18$, $R^2 = 0.83$	Tang et al. (2003)
CO ₂	Peanut field	IRGA installed in soil	Mo.97 MQ.61 Mo.00 Mo.99 Ma1959	(a)	NSSC, every 2–5 days, non-linear curve fitting	$a = 1.03$, $R^2 = 0.67$, r.a. = +3% $a = 1.58$, $R^2 = 0.66$, r.a. = +58% $a = 1.61$, $R^2 = 0.63$, r.a. = +61% $a = 1.81$, $R^2 = 0.64$, r.a. = +81% $a = 2.59$, $R^2 = 0.62$, r.a. = +62%	Pingintha et al. (2010)
CO ₂	Pine stand	IRGA installed in soil	MQ.61 Mo.99	(a)	NSSC “static” = no fan, 30 min measurement, linear curve fitting NSSC, biweekly EC, night-time ecosystem respiration	NSSC ~ GM(MQ.61) > GM(Mo.99) ~ NSSC “static” GM(Mo.99): $a = 1.09$, $R^2 = 0.81$	Sullivan et al. (2010)
CO ₂	Larch plantation	IRGA installed in soil	D_5 -SWC relationship determined	(b)	Automated SSC, hourly Manual SSC, monthly Manual NSSC, monthly	$a = 1.2$, $R^2 = 0.77$; r.a. = +45% r.a. = ca. +50% ^c r.a. = ca. –10% ^c	Liang et al. (2004)
CO ₂	Semi-arid grassland	IRGA installed in soil	Mo.99	(a)	Automated NSSC, non-linear curve fitting Manual NSSC, every 2–4 weeks	$a = 1.08$, $R^2 = 0.71$ $a = 0.95$, $R^2 = 0.82$	Myklebust et al. (2008)
CO ₂	Prairie, fallow	Sampling from gas wells	Measured	(a)	AAC NSSC	GM ~ AAC > NSSC	De Jong et al. (1979)
CO ₂	Larch plantation	IRGA installed in soil	Cu.60 fit to measured D_5	(b)	Automated SSC	r.a. = +12%	Liang et al. (2010)
CO ₂	Douglas fir stand	IRGA installed in soil	Cu.60 fit to measured D_5	(d)	Automated NSSC, linear curve fitting	$a = 0.88$, $R^2 = 0.96$ Rain data excluded: $a = 0.95$, $R^2 = 0.99$	Jassal et al. (2005)
CO ₂	Semiarid steppe, 2 sites	Active Sampling	Tr.82 fit to measured D_5	(a)	NSSC automated, 60 min measurement, linear curve fitting	$a = 1.08$ (1.39), $R^2 = 0.88$ (0.81) r.a. = 3% (60%)	Wolf et al. (2011)
CO ₂	Agricultural field	Gas well sampling	Measured D_5	(a)	NSSC, weekly, 1 gas sample after 15 min	$a = 0.62$, $R^2 = 0.24$	Kusa et al. (2008)
CO ₂	Subalpine forest	IRGA installed in soil	Mo.99	(c)	NSSC, non-linear curve fitting sites 1/site 2	All data: $a = ca. 0.7^c$, $R^2 = 0.51$ Rain + wet soil data excluded $a = ca.$ 0.95/1.3 ^c , $R^2 = 0.89/0.80$	Riveros-Iregui et al. (2008)

Table 1 (Continued)

Gas	Site	Gradient method ^a			Reference system (relevant available details)	Relative accuracy	Reference
		Set-up	D_5 profile ^b	Calculation			
CO ₂	Pine plantation	IRGA in soil	Mo_99	(c)	NSSC, monthly	$\alpha = 1.01$, $R^2 = 0.86$	Tang et al. (2005b)
CO ₂	Mixed temperate forest	IRGA installed in soil	Mo_99	(c)	NSS (2 sites)	$\alpha = 0.99$, $R^2 = 0.73$ $\alpha = 0.90$, $R^2 = 0.91$	Vargas and Allen (2008c)
CO ₂	Tropical forest	IRGA in soil	Mo_99	(c)	NSS	$\alpha < 1$, $R^2 = 0.81$	Vargas and Allen (2008a)
CO ₂	Mixed temperate forest	IRGA installed in soil	Mo_99	(c)	NSS	$\alpha = 1.01$, $R^2 = 0.94$	Vargas and Allen (2008b)
CO ₂	Pine plantation (2 sites)	Sampling from gas wells	MQ_61	Analytical	NSSC, 2 min measurement, linear curve fitting	$\alpha = 1.36$, $R^2 = 0.54$	Suwa et al. (2004)
				Numerical		$\alpha = 1.55$, $R^2 = 0.51$ $\alpha = 1.53$, $R^2 = 0.67$ $\alpha = 1.69$, $R^2 = 0.63$	
CO ₂	Prairie + temperate forest	Sampling from gas wells	MS_71	(a)	NSSC, 45 min measurement, linear curve fitting	Slope $1.1/R^2 = 0.99$	Sanderman and Amundson (2010)
CO ₂	Soil columns	IRGA (in soil)	Mo_00	(a)	NSSC	$\alpha < 1$, $R^2 = 0.72$	Turcu et al. (2005)
CO ₂	Agricultural field	Gas well sampling	Measured D_5	(a)	NSSC, weekly, 1 gas sample after 15 min	$\alpha = 0.62$, $R^2 = 0.24$	Kusa et al. (2008)
N ₂ O	Agricultural field	Gas well sampling	Measured D_5	(a)	NSSC, weekly, 1 gas sample after 15 min	$\alpha = 0.61$, $R^2 = 0.29$	Kusa et al. (2008)
N ₂ O	Semiarid steppe, 2.sites	Active Sampling	Tr.82 fit to measured D_5	(a)	NSSC automated, 60 min measurement, linear curve fitting	$\alpha = 0.28$ (0.44), $R^2 = 0.71$ (0.51), r.a. = –51% (11%)	Wolf et al. (2011)
CH ₄	Boreal forest snowpack	Sampling from gas wells	Mi_59	(a)	NSS, every 10 d, closed for 40 min/4 samples	$\alpha = 0.67$, $R^2 = 0.69$ (CH ₄ production)	Kim et al. (2007)
CH ₄	Semiarid steppe, 2 sites	Active Sampling	Tr.82 fit to measured D_5	(a)	NSSC automated, 60 min measurement, linear curve fitting	$\alpha = 1.44$ (1.81), $R^2 = 0.65$ (0.80), r.a. = 15% (76%) (mainly CH ₄ consumption)	Wolf et al. (2011)
CO ₂	Boreal forest, snowpack	Sampling from gas wells	Mi_59	(a)	NSSC, 40 min measurement, linear curve fitting	$\alpha = 1.2$, $R^2 = 0.69$	Kim et al. (2007)
CO ₂	Grassland snow, straw	IRGA installed in soil	Mo_99	(a)	Automated NSSC, non-linear curve fitting	$\alpha = 1.36$, $R^2 = 0.0$	Myklebust et al. (2008)

^a Different calculation schemes were used: (a) discrete differences (De Jong and Schappert, 1972); (b) Hirano et al. (2003), (c) Tang et al. (2005a); (d) Jassal et al. (2005).^b Different D_5 models were used: Mi_59 (Millington, 1959), MQ_61 (Millington and Quirk, 1961), Ma_59 (Marshall, 1959), Mo_97 (Moldrup et al., 1997), Mo_00 (Moldrup et al., 2000), Mo_99 (Moldrup et al., 1999), MS_71 (Millington and Shearer, 1971), Cu_60 (Currie, 1960), Tr_82 (Troeh et al., 1982).^c Values were estimated from data-plots.

from chamber and GM measurements. But the flux rates were clearly different during windy periods when gas transport within the snow layers and chamber measurements were probably both affected (Takle et al., 2004). Thus, it seems that both the GM and chamber methods are challenging to use in snow.

7. Applicability, limitations and challenges

The studies above show different levels of agreement between the GM and chamber measurements due to several reasons. Assumptions and simplifications have to be made to apply the GM, namely that soil gas transport can be represented by a one dimensional description of Fick's law. This is not always justified.

7.1. Applicability of Fick's law

Using the GM can be problematic in soils near water saturation. Application of Fick's law to describe gas diffusion is challenged by non-equimolar gas fluxes and low relative air-filled pore-volume (Jaynes and Rogowski, 1983; Thorstenson and Pollock, 1989). High soil moisture contents can be associated with locally anaerobic conditions including non-equimolar net balance of gas; CH_4 production, which is usually accompanied by ebullition and plant mediated transport (Fig. 5, left), is one such condition. Hendriks et al. (2010) observed that GM derived flux estimates of CH_4 production were comparable to chamber measurements when the ground water level was low and roots did not reach the water table. At sites with higher water levels or plant roots reaching the ground water table, the GM underestimates the chamber measured flux by an order of magnitude because ebullition and plant mediated transport were the dominating transport process for CH_4 .

An additional practical problem for applying the GM is that the relative error in estimating the low diffusivity values using different D_s models can become very large near water saturation (Rolston et al., 1976; Sallam et al., 1984). Hence, the applicability of the GM is questionable in soils near water saturation.

Even though soil gas transport is dominated by diffusion, mass air flow can occur in certain situations. Air flow due to wind is an important factor for gas transport within the highly porous litter layer (Novak et al., 2000b, 2000a). Large changes in barometric pressure occurring during the passage of frontal systems can cause mass air flow in the soil, especially in deeply aerated soils, fractured unsaturated media (Clements and Wilkening, 1974; Neeper, 2001; Nilson et al., 1991) or special geological structures (Elberling et al., 1998). Changes in the ground water level can result in subsurface pressure variations, causing mass flow of air (Jiao and Li, 2004; Maier et al., 2010). Intense rainfall and flooding may also affect soil gas transport in a similar way. Fortunately, most of these factors affect soil gas transport only during short periods (barometric pressure changes, changes in ground water, intense rainfall) or at special sites (tidal flats).

Wind can induce quasi-static pressure fields across topographical features, causing mass flow in porous media such as snow, soil or rock piles (Amos et al., 2009; Bowling and Massman, 2011). Mass flow of air theoretically can be included in the calculations if the air permeability and the pressure gradients are known. Yet, significant flow rates can be driven by hardly detectable pressure gradients if the air permeability is high (Maier et al., 2010). An alternative approach was developed by Bowling and Massman (2011), who used the $^{12}\text{CO}_2/^{13}\text{CO}_2$ composition of air in a snowpack to quantify the contribution of non-diffusive gas transport. They could show that wind increased the CO_2 transport in the snowpack by up to 40% in the short term and about 10% in the long term.

There are also high-frequency pressure fluctuations ($>0.1\text{ Hz}$) due to wind that affect soil gas transport. Turbulence-driven

pressure-pumping (Massman et al., 1997; Takle et al., 2004) can strongly influence soil gas concentrations (Drewitt and Warland, 2007; Flechard et al., 2007; Hirsch et al., 2004; Maier et al., 2010; Reicosky et al., 2008; Seok et al., 2009). The pumping effect induces an oscillating flow in the air-filled pores that seems to 'enhance diffusion' of the soil gas (Maier et al., 2012). This effect can be attributed to dispersion processes (Kimball and Lemon, 1971b; Maier et al., 2012; Scotter et al., 1967; Takle et al., 2004). The increase in gas transport due to pressure pumping can reach up to 60% of the molecular diffusion in media with coarse pores (Bowling and Massman, 2011; Maier et al., 2012) or even substantially higher if only the top 2 cm of a soil is considered (Kimball and Lemon, 1971a). Gas transport through snowpack seems to be particularly susceptible to pressure-pumping and mass flow (Bowling and Massman, 2011; Bowling et al., 2009; Massman and Frank, 2006; Seok et al., 2009) as well as gas transport in soils with a high air-filled pore-volume (Maier et al., 2012). Massman (2006) and Poulsen and Moldrup (2006) included the transport process of pressure-pumping induced dispersion in gas transport modeling. Poulsen and Moldrup (2006) concluded that wind-induced gas transport can be relevant even for soils with higher soil water content.

Consequently, it seems that increasing wind intensity is associated with an increasing uncertainty in GM derived flux estimates (unless this effect is quantified and included), particularly in snowpack and other media with coarse pores. Monitoring of wind parameters during GM measurement campaigns might allow assessing qualitatively the uncertainty of the flux estimates.

7.2. Spatial variability of soil gas diffusivity and gas production and consumption

Horizontal homogeneity is a second major assumption of the GM that reduces the calculations to a simple vertical 1D description of the gas transport. However, gas production, D_s and subsurface gas concentrations can exhibit high spatial variability (Bekele et al., 2007; Kursar, 1989). Soil compaction in skid trails, for instance, can result in high, small scale variability of D_s and soil CO_2 concentration (Schack-Kirchner et al., 1993), causing lateral diffusion in the soil (Fig. 5 middle). Soil gas diffusivity can vary substantially even in an apparently homogeneous soil (Kühne et al., 2012; Lange et al., 2009). Soil gas diffusivity is anisotropic and the vertical direction is preferential (Kühne et al., 2012). Nevertheless, lateral gas transport can be important in heterogeneous soils and trenching experiments (Jassal and Black, 2006).

Spatial variability of D_s results from heterogeneity in soil pore structure and soil water content. Several soil cores should therefore be sampled at depth to get a reliable mean for the total porosity and diffusivity. In the same way, more than one soil moisture probe at every depth should be used to get a reliable estimate of the mean soil water content (Davidson and Trumbore, 1995; Wolf et al., 2011) because a single probe can under- or overestimate the mean content.

Several authors observed that surface flux estimates derived using the GM were less reliable during rain (DeSutter et al., 2008; Riveros-Iregui et al., 2008; Tang et al., 2005b; Turcu et al., 2005). This deviation can be explained by the flush of microbial activity stimulated in the litter layer, which hardly affects the soil gas concentration, leading to an underestimation of the GM derived flux. An overestimation of the GM, on the other hand, can be explained by a thin top soil layer where the pores are blocked by rain water, thus controlling soil gas exchange. Yet, the increase in soil water content is hardly detectable for regular probes which integrate over a larger volume for the measurement. Hence, accurate estimates of D_s hinge on measurements of soil water content with a high spatial resolution (Turcu et al., 2005). DeSutter et al. (2008) proposed

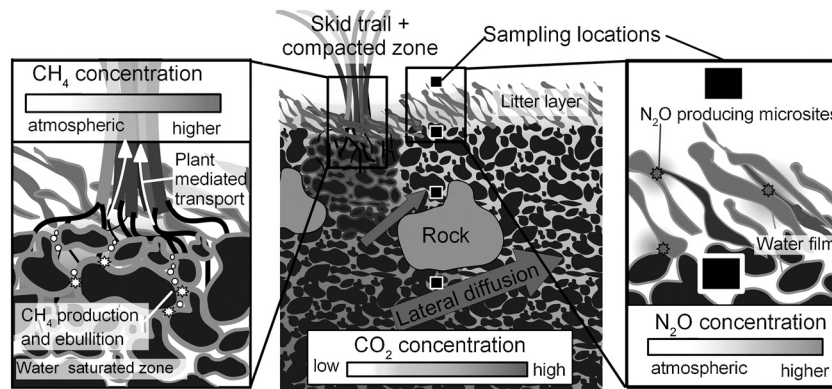


Fig. 5. Conceptualized processes limiting the accuracy of gradient method flux estimates. Middle: Horizontally variable D_s due to compacted zones or large rocks cause lateral diffusion. Left: CH_4 production at sites near water saturation is accompanied by ebullition and plant mediated transport. Right: N_2O production at micro-sites in the litter layer is not reflected in the concentration gradient at the sampling locations.

to use dual-probe heat capacity sensors to measure the soil water content just below the soil surface. Turcu et al. (2005) proposed to model numerically water flow to derive the profile of soil water contents in the top few centimeters.

7.3. Assumption of quasi steady-state diffusion

Quasi steady-state diffusion is assumed in most studies. However, soil gas concentrations change with time. Such changes might seem negligible over periods of several days or weeks (De Jong and Schappert, 1972; Schwendenmann et al., 2003), although they can be important on the scale of hours (De Jong and Schappert, 1972; Maier et al., 2010). Whether the simplification by assuming a steady-state (i.e. the neglect of a storage flux) is justified depends on the process investigated.

The response of soil respiration to rain ranges from less than hours to days. Rainwater can stimulate soil respiration, increasing soil CO_2 production. At the same time rainwater blocks pores in the top soil, impeding gas transport and decreasing D_s . Both effects lead to increasing soil CO_2 concentrations. This change in the amount of stored CO_2 can be rather small compared to the efflux, for instance at acidic sites (Jassal et al., 2005). But the change rate can be substantial in deeply aerated, calcareous soils, reaching up to 60% of the efflux and, thus, leading to an important difference between soil CO_2 efflux and CO_2 production (Maier et al., 2010, 2011). It should be noted that the soil solution can be an important storage for CO_2 in soils with a high pH (Flechar et al., 2007; Maier et al., 2010).

Contrary to the expected CO_2 increase after rainfall, short, temporary declines in soil CO_2 concentration were observed in the top soil at calcareous sites (Maier et al., 2010; Myklebust et al., 2008). This rarely observed phenomenon was explained by the dissolution of soil CO_2 in the soil solution, which was under-saturated in CO_2 due to mixing with rainwater. Non-steady state situations were also observed after the removal of snow (Elberling and Ladegaard-Pedersen, 2005) and in thawing soil when trapped CO_2 caused an unexpected high flux (Elberling, 2003).

The role of a transient storage has been scarcely discussed for gases other than CO_2 . Wolf et al. (2011) observed vigorous N_2O production in a top soil layer near water saturation, and pointed out that this is most likely evidence against quasi steady-state conditions.

7.4. Processes and gases

A good agreement between GM and chamber measurements of CO_2 emissions was found in most studies (Table 1). This agreement

seems to be stronger for CO_2 efflux and the net consumption of CH_4 than for efflux of N_2O and CH_4 (Dunfield et al., 1995; Jassal et al., 2005; Kim et al., 2007; Pihlatie et al., 2007; Vargas and Allen, 2008a; Wolf et al., 2011; Yavitt et al., 1995).

Spatial and temporal variability of N_2O and CH_4 fluxes can be very high and range from net consumption to (higher) net production (Ambus and Christensen, 1995; Wolf et al., 2011; Hendriks et al., 2010). Chamber measurements can only provide information about the horizontal variability of soil–atmosphere fluxes of N_2O and CH_4 , while the GM allows additional assessment of the vertical dimension of sources and sinks in the soil profile. N_2O and CH_4 are produced under anaerobic conditions, which can occur in soil compartments near water saturation (e.g. near the ground water table) but also in anaerobic micro-sites (Butterbach-Bahl and Papen, 2002; Hendriks et al., 2010; Seiler and Conrad, 1981). Anaerobic micro-sites can occur even within a generally aerobic soil (Burford and Stefanson, 1973). This means CH_4 and N_2O can be produced at micro-sites next to where they can be consumed. As a result, concentrations of N_2O and CH_4 can vary on a small scale in a soil profile as shown in Fig. 5.

When applying the GM, the soil gas profile is usually sampled at (more or less) discrete locations. The true concentration gradient at these locations remains unknown (Fig. 5). The gradient can be reliably estimated if the gas production or consumption is evenly distributed between two or more measurement locations. If the production or consumption varies substantially at smaller scales, and consequently the gas concentration too, then the estimated gradient includes large uncertainties. This is especially problematic in the layer above the uppermost gas sampling location because the surface gradient and the derived flux will be poorly estimated (Hashimoto and Komatsu, 2006). The production of N_2O and CH_4 in anaerobic micro-sites in the litter layer is an example of such small scale processes (Fig. 5, left, right).

Differences in flux estimates of N_2O and CH_4 derived using the GM and chamber methods are often attributed to processes in the top soil layer above the uppermost sample (Arah et al., 1991; Kusa et al., 2008; Maljanen et al., 2003; Pihlatie et al., 2007; Wolf et al., 2011). Maljanen et al. (2003) explained the lower GM estimates by high N_2O production in the wet topsoil above the shallowest sampling location. Pihlatie et al. (2007) attributed overestimation to possible consumption of N_2O above the top soil-gas sampler. The same issue was addressed by Wolf et al. (2011) who found higher GM derived estimates of CH_4 consumption than chamber derived estimates. They concluded that some CH_4 production occurred close to the surface. On the other hand, it seems that the soil–atmosphere flux of CH_4 can be reliably determined using the GM if the production occurs only in the deeper soil and gas transport

in the overlying soil corresponds to molecular diffusion (Hendriks et al., 2010).

The GM is an appropriate method to estimate net CH_4 consumption if CH_4 production can be excluded (Potter et al., 1996; Dörr et al., 1993). Consumption of CH_4 apparently occurs ubiquitously in aerated upland soils (Le Mer and Roger, 2001), and seems to be limited by diffusion of atmospheric CH_4 into the soil at numerous sites (Potter et al., 1996; Dörr et al., 1993). Similar to the consumption of CH_4 in a well aerated soil, the production of CO_2 stretches across the soil profile and can be easily resolved compared to anaerobic microsites. The distribution of roots and organic matter can give a rough idea of the most active respiration zones, meaning it is possible to adapt the design of the gas sampling system beforehand. Respiration processes in the litter layer can become important, for instance when rain falls onto dry litter. The rain stimulates a flush of microbial activity and CO_2 emissions (Lee et al., 2004; Liu et al., 2002; Xu, 2004), but the gas concentration in the soil below is hardly affected (Tang et al., 2005a) and the soil–atmosphere flux is poorly estimated using the GM. Yet, this problem occurs only during short periods (Jassal et al., 2005; Riveros-Iregui et al., 2008). Generally, estimation of CO_2 fluxes using the GM and chamber methods agree well. In the recent decade, several studies using solid-state CO_2 sensors showed even stronger agreement between the near-continuous GM measurements and chamber derived flux estimates, making the GM a suitable tool for near-continuous flux monitoring.

8. Recommendations

We think that the GM is a suitable method to assess fluxes of CO_2 and the consumption of CH_4 in soils. It can also be used to monitor the production of N_2O and CH_4 , but this should be accompanied by additional chamber measurements. The chamber measurements are necessary to reduce the uncertainty arising from possible N_2O and CH_4 production in the top-layer, unless such a production can be ruled out (e.g. dry conditions, aeration). We recommend using the GM in soils near water saturation very carefully, because uncertainty in D_s estimation is particularly high and transport processes such as ebullition and plant mediated transport may play a role. The profile for the application of the GM should be carefully chosen; profiles that exhibit obvious, horizontal inhomogeneity (e.g. Fig. 5) should be avoided.

We recommend adapting the design of gas sampling system to the focus of the study. If the focus is on estimation of the soil–atmosphere flux the gas sampling locations should be as close to the surface as possible. Yet, accuracy of estimation of depth and D_s must be guaranteed. A sensible compromise follows the approaches of Tang et al. (2005b) or Hirano et al. (2003). If the gas concentration at the soil surface is needed for the calculation of the soil–atmosphere flux, the soil surface concentration should be monitored, especially in soil with a high D_s and low concentration gradient. If the focus is on vertical partitioning, several sampling locations should be distributed across the zone that includes the largest changes of gas production or consumption. More sampling locations allow for a higher vertical resolution. If the installation of the GM system (gas sampling + soil moisture probes) causes major soil disturbance (e.g. the excavation and refilling of a pit), sufficient time has to be allocated before starting the soil gas measurements.

Since estimation of the soil gas diffusivity is a significant source of uncertainty, we recommend including D_s measurements, which is only a little extra work in view of the dimension of most studies. Special attention should be given to the litter layer, since data on diffusivity are scarce. Simple one-chamber methods for the measurement of D_s on undisturbed soil cores are described by Jassal et al. (2005) and Kühne et al. (2012). Soil water content directly affects the estimation of D_s . Therefore soil water content should be

measured with several replications at every sampling depths and with a good vertical resolution in the top soil. A soil specific calibration of the soil moisture probes is also generally recommended. It can be advantageous to couple measurements with modeling soil water flow to interpolate the vertical profile of soil water content. Alternatively, *in situ* determination of soil gas diffusivity like the Rn method has shown to yield good results.

Reasonable calculation approaches are available to calculate the soil–atmosphere flux of CO_2 (De Jong and Schappert, 1972; Hirano et al., 2003; Tang et al., 2005b; Jassal et al., 2005). The assumption of homogeneous gas production in the top layer of the soil profile, which is implicit in some approaches, may not be true for all gases. Hence, for gases other than CO_2 only the approach of De Jong and Schappert (1972) is recommended. Fitting empirically analytical functions to discrete depth profiles of gas concentrations so to derive a continuous profile should not be done to calculate the soil–atmosphere flux. For vertical partitioning, the choice of an analytical function to fit a continuous concentration profile or to describe the production profile must be carefully considered and justified, or avoided. The assumption of quasi steady-state diffusion is not always true for soil CO_2 fluxes. We recommend including storage changes at sites with a large CO_2 storage capacity (deeply aerated soils with high pH), especially after rainfalls. An alternative, promising approach is to include measured data in numerical modeling of soil gas fluxes. Yet, as with analytical approaches, it is important that the assumptions about the vertical profiles of gas production or consumption and of D_s are met when including numerical evidence.

Non-diffusive gas transport in soil can be an important problem at some sites for some time. For instance, in water saturated soils ebullition may be the dominating transport process for methane. Turbulence-driven pressure-pumping can affect soil gas transport at many sites even over longer periods. This process should be accounted for, especially in long term studies and in soils with a high air-filled pore-space and snowpack. Bowling and Massman (2011) developed a method to separate non-diffusive and diffusive transport in porous media that could help to quantify and better understand gas transport in soil and snowpack.

9. Conclusions

Ideally, the GM should be used on well aerated, horizontally homogeneous soils where gas exchange is entirely driven by diffusion. Here the GM promises to yield reliable results when soil respiration and methane consumption is studied (Fig. 1). Substantial discrepancy in these conditions can lead to increasing uncertainty in the flux estimates. Soils near water saturation are particularly problematic because the estimation of D_s includes a higher uncertainty and other gas transport processes, such as ebullition and plant-mediated gas transport, might dominate. Gas transport in well-aerated soils with low soil water content or snowpack, on the other hand, seems to be more susceptible to turbulence-driven pressure-pumping during periods of high wind intensity. Being aware of these uncertainties, the GM can be applied across a wide range of sites and soil moisture. Using the GM to quantify the production of compounds such as N_2O , which are produced in “hot spots”, or CH_4 must be carefully considered, especially if the production occurs in the organic layer. For this purpose, the GM application should be amended with chamber measurements.

The GM complements and represents an alternative to chamber and eddy covariance methods to estimate gas fluxes from soils. The laborious field installation and need to measure a considerable set of soil physical parameters limit the mobility of the GM application. Possible near-continuous measurements and the minimally affected soil surface environment (no impact on precipitation or

litter fall) make the GM a suitable method for intensive short term and long term monitoring of soil gas fluxes. Additionally, the GM can provide information about the depth profile of gas production or consumption in the soil, making it a unique tool for many field studies.

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